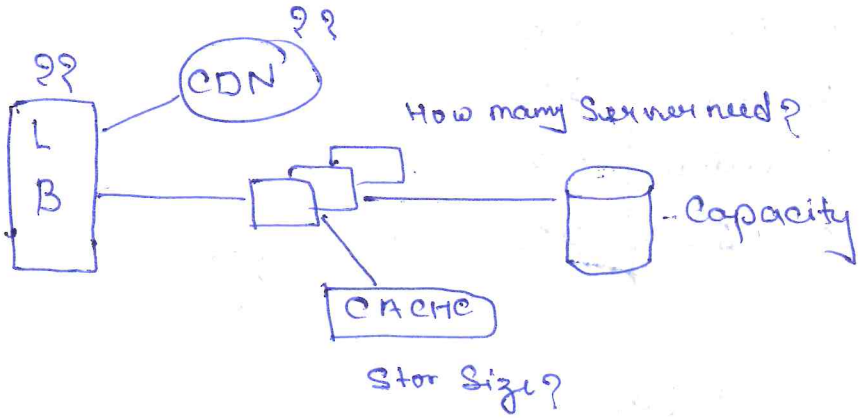
Back-of-The-Envelope Estimation

What is Back-of-The Envelope Estimation?



Need

Drives of decision for System Design.

Considaration

→ Rough (T-shirt Size) Estimation

→ Do Not spend much time?

→ Keep the assumption value simple.

275 mil x 10 M

435 mil x 100 m

1000 m

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Cheat Sheet *** - Every things is multiple of 3 zeros

~~3 zeros~~	Traffic	Storage
3 zero	Thousend	KB
6 zero	Million	MB
9 zero	Billion	GB
12 zero	Trillion	TB
15 zero	Quadrillion	PB — Peta Byte.

you have come with data.

(1 byte) ASCII — (2 byte) UNICODE

Char → 2 bytes

Long/double → 8 bytes

Integer — 16 bytes

Average Image = 300 KB

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→ No of Servers ?

→ RAM ?

→ storage Capcity ?

Trade off (CAP)

X Million User * y MB ⇒ XY TB Storage
6 zero (6 zero)

5 Millio User * 2 KB ⇒ 10 GB
6 zero 3 9 zero

Estimation of Facebook :-

1 Traffic Estimation :

Total User : 1 Billion

Daily Active User :- 25% of total User = 250 Million

* We are saying daily active users are 250 million users so can we come up with an estimate like daily how many request query we can get

Read operation & write Operation

5 + 2 = 7 Query

⇒ (250 Million Users X 7 query) / 100000 second.
(DAU)

60 X 60 X 24 ≈ 86400 Second

⇒ 18 K query per Second

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(2) Storage Assumption :-

(1) ⇒ Every User (D.A.U) doing 2 post (Each Post 250 character)

(2) ⇒ 10% of User 1 Image. (300 KB Size)

(1) Post: 1 Post → 250 Charater →

250 * 2 = 500 byte
1 byte

2 Post → 500 x 2 = 1000 Byte ⇒ 1 KB

250 Million User * 1KB = 250 GB/day ?   (6 zero) + (3 zero)

(2) Image :-

25 Million Users * 300 KB = 7500 GB
(6 zero) (3 zero)

7000 GB

7 x 1000 GB

~ 8 TB

~ 8 TB / day

to store the data 5 years

5 years = 1825 day ~ 2000 days.

(Post) 2000 * 250 GB = ~ 500 TB

(Image) 2000 x 8 TB ⇒ ~ 16 PB

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3) RAM Estimation

for Each User (DAU) Last 5 posts we are puting into Cache.

1 Post = 500 Byte

5 * 500 = 2500 bytes

250 Million (DAU) * 2500 bytes / 3000

250 Mill (6 zero) 3 K (3) = 750 GB

Memory space

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1 Machine = 75 GB hold in Memory Data

10 Machine for Caching (Nodes)

Latency :→ 95% 500 ms 1 request

18 K req/second 1 Second 2 (500 ms)

~~1 Machine →~~

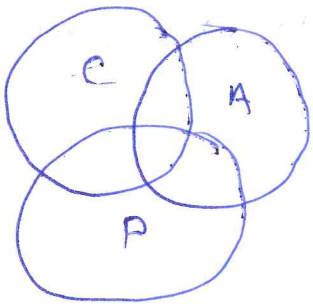
1 Server → 50 threads ⇒ 100 request/Second

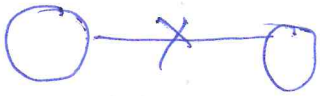
1 sec → 2 req

How many server = 180 Server

CAP Theorem trade off

CA
CP
AP





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— This topic is not fully clear. — Any time Practice.